Reflexivity of natural deduction derivability

The Open Logic Project contributors

openlogic:first-order-logic/natural-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author proposition labelled prop:reflexivity (771-774); the proof span 776-779 is the author proof environment that immediately follows it.

Editorial note. No cross-references occur in the retained span.

Editorial scope. One original result of the Open Logic Project unit first-order-logic/natural-deduction, counted once: reflexivity of derivability. Only the counted proposition and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Derivability from assumptions. Gamma proves !A when there is a derivation of !A in which every undischarged assumption is a sentence in Gamma.. It is the relation the counted proposition and proof operate on (771-779).

Proposition 0.1 (Reflexivity)**.** *If* $!A \in \Gamma$*, then* $\Gamma \vdash !A$*.*

Proof. The assumption $!A$ by itself is !!aderivation of $!A$ where every !!undischarged assumption (i.e., $!A$) is in Γ. □

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